Background

History, properties, geology and reserves of lithium

Lithium is the lightest and least dense solid element in the periodic table with a standard atomic weight of 6.94. In its metallic form, lithium is a soft silvery-grey metal, with good heat and electric conductivity. Although being the least reactive of the alkali metals, lithium reacts readily with air, burning with a white flame at temperatures above 200 °C and at room temperature forming a red-purple coating of lithium nitride. In water, metallic lithium reacts to form lithium hydroxide and hydrogen. As a result of its reactive properties, lithium does not occur naturally in its pure elemental metallic form, instead occurring within minerals and salts.

Properties of lithium

Atomic number	3
Atomic weight	6.94
Melting point (°C)	180.54
Boiling point (°C)	1347.0
Density (g/cm^3)	0.53
Oxidation states (valence)	+1
Colour	Silvery-grey
Thermal conductivity (Wm-1K-1)	84.8
Electrical resistivity (nΩ/m)	92.8
Stable isotopes	Li-6, Li-7

Source: Chemistry Daily

A. Białobrzeski et. al., Ultralight Magnesium-Lithium Alloys, Archive of Foundry Engineering, Vol. 7 Issue 3, 2007

Notes: Wm-1K-1 = Watts per metre kelvin
nΩ/m = nano Ohm per metre

Occurrence of lithium

The crustal abundance of lithium is calculated to be 0.002% (20 ppm), making it the 32nd most abundant crustal element. Typical values of lithium in the main rock types are 1-35 ppm in igneous rocks, 8 ppm in carbonate rocks and 70 ppm in shales and clays. The concentration of lithium in seawater is significantly less than the crustal abundance, ranging between 0.14 ppm and 0.25 ppm.

There are five naturally occurring sources of lithium, of which the most developed are lithium pegmatites and continental lithium brines. Other sources of lithium include oilfield brines, geothermal brines and clays.

Lithium minerals

In comparison to many of the other cations that occur in magmas, lithium preferentially remains in the fluid phase of the melt. As a result of this higher solubility, lithium minerals occur mainly within late-stage intrusives such as pegmatites and alkaline intrusives. Lithium minerals also occur within hydrothermal deposits that have been derived from magmatic fluids.

There are around 250 identified lithium-bearing minerals, although many of these only contain minor amounts of lithium in their composition. According to Garrett, 145 minerals contain lithium as a major component, with 25 containing over 2% Li_2O.

Spodumene is the most commonly mined mineral for lithium, with historical and active deposits exploited in China, Australia, Brazil, the US and Russia. The high lithium content of spodumene (8% Li_2O) and well-defined extraction process, along with the fact that spodumene typically occurs in larger pegmatite deposits, makes it an important mineral in the lithium industry.

Lepidolite is a monoclinic mica group mineral typically associated with granite pegmatites, containing approximately 7% Li_2O. Historically, lepidolite was the most widely extracted mineral for lithium; however, its significant fluorine content made the mineral unattractive in comparison to other lithium-bearing silicates. Lepidolite mineral concentrates are produced largely in China and Portugal, either for direct use in the ceramics industry or conversion to lithium compounds.

Petalite contains comparatively less lithium than both lepidolite and spodumene, with approximately 4.5% Li_2O. Like the two aforementioned lithium minerals, petalite occurs associated with granite pegmatites and is extracted for processing into downstream lithium products or for direct use in the glass and ceramics industry. Upon heating, petalite converts to beta spodumene-quartz in the solid solution phase.

Zinnwaldite is a lithium-bearing mineral of the mica group, named after the town of Zinnwald-Georgenfeld on the German-Czech border. The mineral occurs associated with griesens, pegmatites and hydrothermal veins, and has historically been linked with tin mineralisation. Commercial production of zinnwaldite for lithium has not been achieved, though projects based on zinnwaldite mineralisation are under development.

Lithium minerals such as amblygonite and eucryptite occur in more minor amounts, accessory to the more common spodumene, petalite and lepidolite minerals. Inclusions of these minerals, however, can significantly increase the lithium content of the rock mass.

Significant lithium minerals

Name	Formula	Theoretical maximum Li2O content (%)	Average Li2O% of ores
Spodumene	LiAlSi2O6	8.0	2.9-7.7
Petalite	LiAl(Si4O10)	4.5	3.0-4.7
Lepidolite	K(Li,Al)3(Si,Al)4O10(OH,F)2	7.7	3.0-4.1
Amblygonite	(Li,Na)Al(PO4)(F,OH)	7.4	…
Montebrasite	LiAl(PO4)(OH,F)	10.2	7.5-9.5
Zinnwaldite	KLiFeAl(AlSi3)O10(OH,F)2	3.4	0.4-0.8
Eucryptite	LiAlSiO4	11.8	4.5-6.5
Bikitaite	LiAlSi2O6H2O	7.3	…
Cookeite	LiAl4 (AlSi3O10)(OH)8	2.9	…
Virgilite	LiAlSi2O6	4.1	…
Jadarite	LiNaSiB3O7OH	7.28	1.75-2.0

Source: webmineral.com, mindat.org, Garrett 2004 – Handbook of lithium and natural calcium carbonate

Lithium clays

Lithium clays are formed by the breakdown of lithium-enriched igneous rock, which may also be enriched further by hydrothermal/metasomatic alteration. The most significant lithium clays are members of the smectite group, in particular the lithium-magnesium-sodium end member hectorite. Hectorite ores typically contain lithium concentrations of 0.24%-0.53% Li and form numerous deposits in the US and northern Mexico. As well as having the potential to be processed into downstream lithium compounds, hectorite is also used directly in aggregate coatings, vitreous enamels, aerosols, adhesives, emulsion paints and grouts. Other lithium-bearing members of the smectite group are detailed in **Error! Reference source not found.**.

Major lithium-bearing smectite group members

Name	Formula	Theoretical Li2O content (%)
Hectorite	$Na_{0.3}(Mg,Li)_3Si_4O_{10}(OH)_2$	1.17
Salitolite	$(Li,Na)Al_3(AlSi_3O_{10})(OH_5)$	1.65
Swinefordite	$Li(Al,Li,Mg)_4((Si,Al)_4O_{10})_2(OH,F)_4nH_2O$	3.76

Source: webmineral.com, mindat.org, Garrett 2004 – Handbook of lithium and natural calcium carbonate

Lithium brines

Lithium-enriched brines occur in three main environments: evaporative saline lakes and salars, geothermal brines and oilfield brines. Evaporative saline lakes and salars are formed as lithium-bearing lithologies, which are weathered by meteoric waters forming a dilute lithium solution. Dilute lithium solutions percolate or flow into lakes and basin environments, which can be enclosed or have an outflow. If lakes and basins form in locations where the evaporation rate is greater than the input of water, lithium and other solutes are concentrated in the solution, as water is removed via evaporation. Concentrated solutions (saline brines) can be retained subterraneously within porous sediments and evaporites or in surface lakes, accumulating over time to form large deposits of saline brines.

Salars and saline lakes are exploited in North and South America, as well as in China, and have provided significant amounts of lithium to the global market since the mid-1990s. Lithium brine extraction from saline lakes and salars was first undertaken commercially in 1966 at the Silver Peak deposit in Nevada, US. Lithium had previously been extracted from the Searles Lake deposits in the US, although only as a by-product of potassium production between 1936 and 1978.

In the late 1960s, exploration by the Chilean Instituto de Investigaciones Geologicas identified elevated lithium concentrates in brines at the Salar de Atacama in northern Chile. Commercial production of lithium carbonate from the Salar de Atacama began in 1986 by Sociedad Chilena de Litio. Livent (formerly FMC) began production of lithium chloride from the Salar de Hombre Muerto in Argentina in 1994, shortly before Sociedad de Quimica (SQM) began production of lithium carbonate at the Salar de Atacama in 1995.

Commercial production of lithium from saline lakes in China began in 2004, with production at the Zabuye salt lake in Tibet and West Taijiner saline lake in Qinghai province. The Qaidam basin area of Qinghai province is the most abundant location for lithium-bearing saline lakes in China, with three companies operating lithium brine facilities in the area. The Zabuye Salt Lake is also of particular interest as it is the only known location where Zabuyelite (Li_2CO_3) is precipitated naturally from saline brines.

The chemistry of saline brines is unique to each deposit, with brines even changing dramatically in composition within the same salar. The overall brine composition is crucial in determining a processing method to extract lithium, as other soluble ions such as magnesium (Mg), sodium (Na) and potassium (K) must be removed during processing.

Brines with a high lithium concentration and low Li:Mg and Li:K ratios are considered most economical to process. Brines with lower lithium contents can be exploited economically if evaporation costs or impurities are low. Lithium concentrations at the Salar de Atacama in Chile and Salar de Hombre Muerto in Argentina are higher than the majority of other locations, although the Zabuye Salt Lake in China has a more favourable Li:Mg ratio.

The potential to produce lithium from geothermal brines has been assessed by a number of exploration and development companies. Controlled Thermal Resources is investigating the production of lithium from waste streams of a geothermal power plant near to the Salton Sea in southern California, US, though no commercial production has been achieved at the project to date. Similarly, Vulcan Energy Resources is assessing extraction of lithium from geothermal brines in Germany and Cornish Lithium is assessing the production of lithium from geothermal brines in Cornwall, UK.

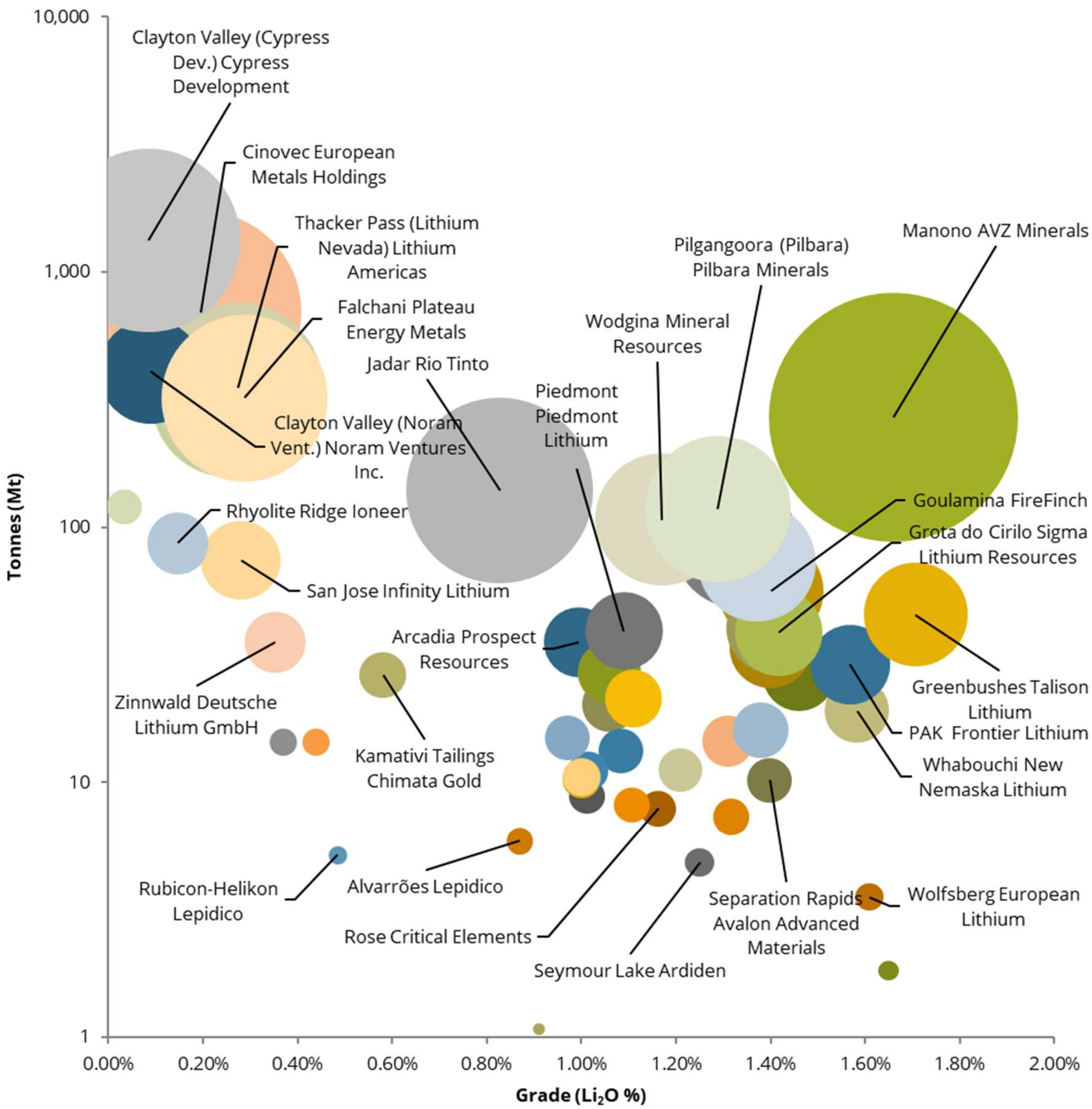
A number of oilfield brines are enriched in lithium, including the Smackover oilfield brines in Arkansas, US, and the oilfield brines in the Devonian Petroleum System in Alberta, Canada. The Smackover oilfield brines are considered to possess the highest lithium concentrations averaging 146-170 ppm Li. They are derived from concentrated seawater enriched in calcium by dolomitisation of oolitic limestone in which the brines are hosted, and enriched in lithium by interaction with geothermal fluids. Albemarle extracts bromine from the Smackover oilfield brines in Arkansas and is developing a method to recover lithium from extracted brines. In Alberta, lithium-enriched brines are reported to occur within porous carbonate source rocks, including the Late Devonian Beaverhill Lake (Swan Hills), the Winterburn (Nisku) formation, the Woodbend (Leduc) formation and the Alberta Basin. Lithium concentrations are reported at above 50 ppm. MGX Minerals and E3 Metals are exploring ways to extract lithium from oilfield brines occurring in the Devonian Petroleum System in Alberta. Besides securing land claims for potential lithium production, they develop technologies to extract lithium from wastewater occurring at oil production before it is reinjected in the ground. MGX Minerals successfully extracted lithium carbonate from heavy oil wastewater in January 2017 and delivered its first petrolithium wastewater treatment system in October 2018.

Reserves and resources

The high grade of Talison Lithium's ore at Greenbushes makes it the largest operational mineral deposit with 6.31 Mt LCE contained, despite the resource size being smaller than Pilbara Minerals' Pilgangoora deposit. The Manono project in the DRC, under development by AVZ Minerals, has the largest identified mineral resources of any mineral project, containing 11.0 Mt LCE. The majority of mineral deposits have grades between 0.8% and 1.6% LCE, with advanced projects differentiated more by their scale.

The largest mineral reserve at a project under development is held by AVZ Minerals at the Manono project in the DRC, with 5.3 Mt LCE reported contained in mineral reserves. The Sonora lithium clay project by Bacanora Minerals is the largest clay project under development with reported reserves containing 4.52 Mt LCE, followed by Covalent Lithium's Mt Holland project in Australia and Lithium Americas' Thacker Pass project in the US with reserves containing 3.49 Mt and 3.14 Mt LCE, respectively.

Mineral resource estimates for lithium mineral and clay deposits, 2021 (Mt vs Li_2O %)



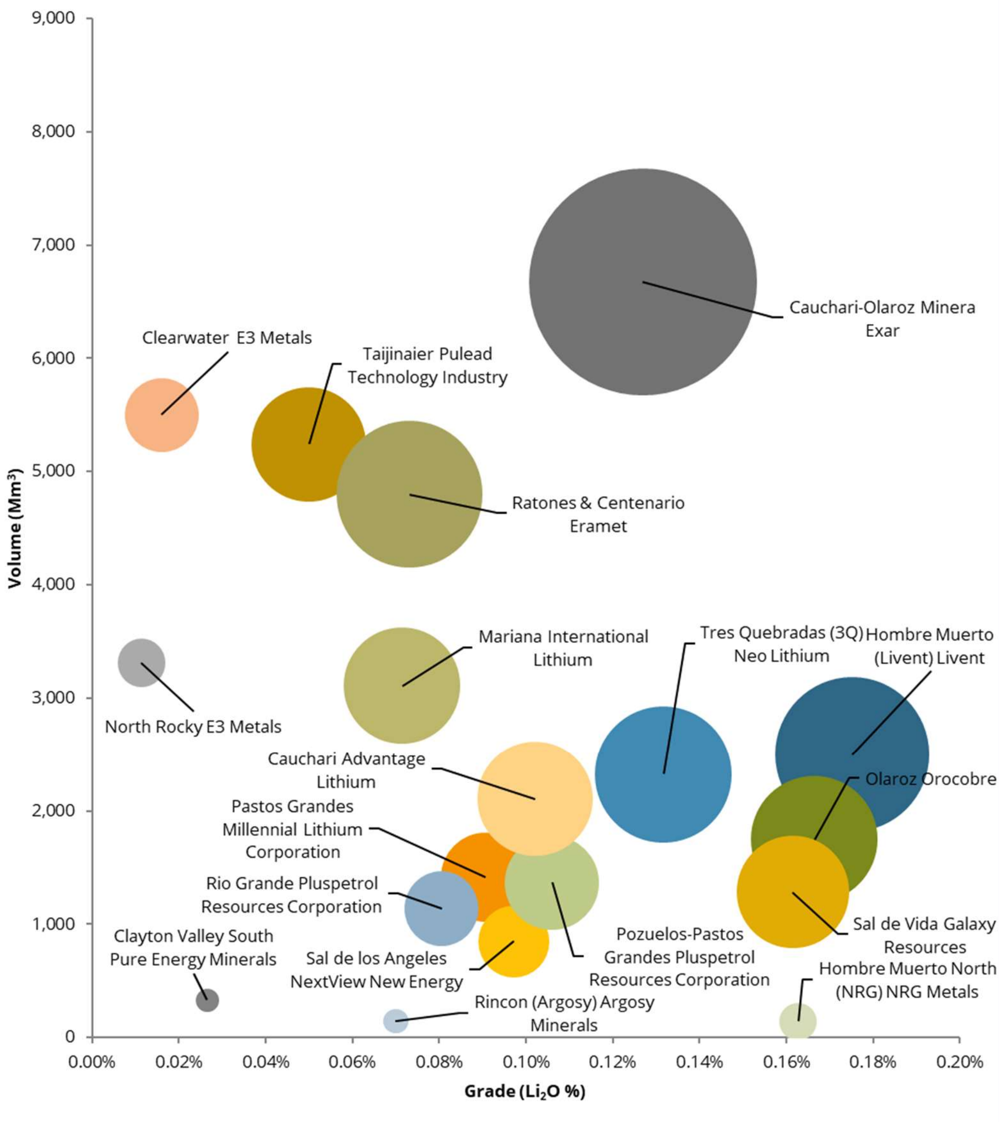
Source: Company data, Wood Mackenzie

Notes: Bubble size represents lithium contained in mineral resources in terms of LCE

Lithium brine and clay mineral resources are generally larger than those of lithium mineral deposits, typically containing 2.0-10.0 Mt LCE, though the grade of lithium brines is much lower at between 0.05% and 0.1% Li. Lithium clay resources reside in the space between lithium brine and mineral deposits in terms of their resource estimate, both in terms of grade and bulk size. However, the lithium content of clay mineral resources is more comparable to lithium brine deposits at 5.0-7.0 Mt LCE.

SQM's Salar de Atacama operation in the Atacama region of Chile has the largest reserves among brine projects, reported at 45.5 Mt LCE . Albemarle's operation located in the same salar is believed to have significant reserves also, though official estimates are not available. Lithium brine projects typically have larger contained lithium volumes in reserves as a result of the large scale of many salt lakes.

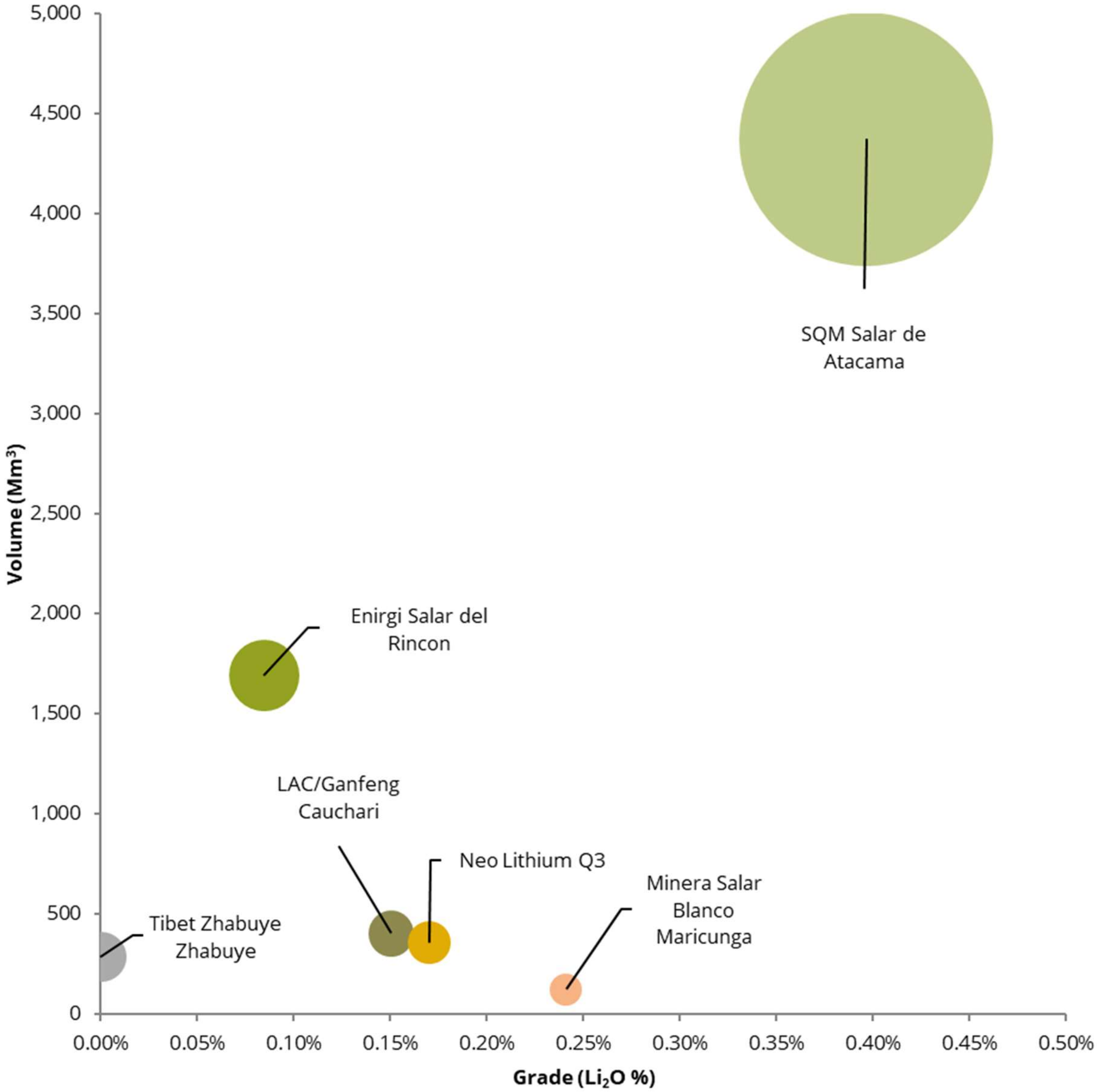
Mineral resource estimates for lithium brine deposits, 2021 (Mm^3 vs Li_2O %)



Source: Company data, Wood Mackenzie

Notes: Bubble size represents lithium contained in mineral resources in terms of LCE

Mineral reserve estimates for lithium mines and advanced projects, brine deposits, 2021 (Mt vs Li_2O %)



Source: Company data, Wood Mackenzie

Mining and Processing

Current lithium extraction is split between two methods: conventional mineral extraction, in which lithium is extracted from mined spodumene-, petalite- or lepidolite-based deposits, and brine, which pumps lithium-bearing brines and concentrates them via either solar evaporation or direct lithium extraction (DLE). Mineral production is often non-integrated, with miners able to sell concentrate to downstream refineries for conversion to a lithium carbonate or hydroxide product, or for use directly within technical applications. However, integration between the mineral concentrate production and mineral conversion stages of the supply chain has become more widespread, with several project reassessing the product entry point in the supply chain to maximise product value add. Brine producers are generally fully integrated given the limited scope for sale of an intermediate product within the process.

Since 2016, lithium supply has shifted away from being dominated by brine supply to one where lithium minerals are the dominant source. This has been the result of several mineral operations coming online in Australia and Brazil looking to take advantage of higher lithium prices during 2017 and 2018. While production curtailments and consolidation of mineral operations has occurred as prices fell back in 2019-2020, production of contained lithium in raw materials has remained above that derived from brine sources.

Mining and extraction

Mineral extraction

Lithium is widespread throughout the Earth's crust, but it does not occur in its natural form due to its high reactivity. Currently, lithium is extracted from three main deposit types – spodumene, petalite and lepidolite, all of which occur in granitic pegmatites. Concentrations of lithium also occur in the form of lithium-bearing clays; however, the extraction of economic volumes from these have so far not been achieved.

Mineral extraction is carried out using conventional mining techniques, analogous to other hard rock mining sectors. The majority of lithium mining is carried out in open pits on spodumene-bearing pegmatites, though there is one currently operational underground mine that uses sub-level stoping. Due to the hard nature of lithium minerals, their extraction is primarily carried out via drill and blast techniques and then excavated from the debris piles for concentration.

Brine

There are two main forms of brine extraction and processing: evaporation and refining, used by SQM, Albemarle and Orocobre, and direct lithium extraction (DLE), used by Livent and a number of Chinese producers. While evaporation relies on solar energy to concentrate brines in evaporation pools, DLE can be used to produce lithium products from high-magnesium and sulphate brines and uses chemical and physical processes to concentrate brine liquor. This process minimises the need for evaporation ponds and the issues surrounding water usage in brine extraction as a large proportion of the depleted brine can be reinjected.

Both processes involve a first phase production of a lithium chloride intermediate. In the evaporation methods employed by SQM, Albemarle and Orocobre, this is done via the gradual staged removal of dissolved salts within the brine via solar evaporation and brine concentration. The brine is pumped into a sequence of shallow ponds and as solar evaporation removes water from the system, salts present within the brine crystallise out of solution from least soluble to most soluble. The sequence of crystallisation will vary depending on the chemistry of the brine, but in general salt precipitate out as follows: halite ($NaCl$), silvinite (KCl), carnallite ($KMgCl3 \cdot 6H2O$), bischofite ($MgCl2 \cdot 6H2O$) and lithium carnallite ($LiCl \cdot MgCl2 \cdot 7H2O$). The result of this is a concentrated lithium chloride brine, typically around 6% Li and 1.8% Mg content.

DLE is less widely used in commercial-scale production, owing to the proprietary technology used in the process, though several projects and operations plan to introduce DLE processing into the flowsheet. In 2021, DLE is used by Livent at its Salar del Hombre Muerto operation in Argentina as well as a number of Chinese operations in Qinghai province. Livent pumps brine to the surface at a concentration of 600 ppm and condition it to the required pH and temperature. Sorption-desorption is then carried out on the brine feedstock, producing a lithium chloride solution. This solution is then subject to evaporation in ponds, under the same principle as operations using the solar evaporation process. The solution is purified and concentrated, with impurities in the forms of salts crystallising out of solution in stages. Depending on the end product required, volumes of solution are pumped out at different concentrations. A 6% Li content solution is used for refined lithium chloride production, while a 3% Li content solution is used for lithium carbonate production.

Mineral processing

Mineral processing can be separated into two stages. Stage 1 is the production of a lithium-bearing concentrate (which can be used directly in technical applications) and stage 2 is the conversion of that concentrate into a refined lithium, hydroxide or carbonate, product.

Concentrate production

The concentrate production flowsheet of a mineral producer will vary depending on the mineralogy and petrology of the deposit, but there are several key steps shared throughout the industry. Many operations will carry out additional steps, such as secondary recovery circuits. The general flowsheet for production of a spodumene concentrate is as follows:

- **Crushing and grinding:** Ore is crushed and milled to lower the grain size of the ore feedstock. Depending on the flowsheet and planned proceeding steps, the feedstocks post-crushing and grinding grain size will vary. Producers will usually employ several crushing and grinding circuits integrated together and typically between two to four circuits with screening are used.
- **Separation:** Coarse-grained material, such as spodumene ore, is amenable to separation via gravity separation and the more common method employed is dense media separation (DMS). In the same vein as crushing and grinding, DMS will be carried out over primary and secondary circuits utilising separate medium liquids to improve overall recoveries. As part of this stage, operations will frequently employ separate separation circuits, such as hydroseparation, to remove mica from the feedstock.
- **Flotation:** Froth flotation is not universally employed within all spodumene concentrate circuits. The stage is used to remove silicate minerals (feldspar and quartz) from the concentrate and further concentrate the ore and is the second concentration step (after DMS) in many spodumene processing plants.

Other steps are frequently taken to improve mineral recovery rates as part of secondary circuits, separate from the main flowsheet. These steps will be taken based on the mineralogy of the deposit and will therefore vary from operation to operation. A common further step, separate from the main circuits, is gravity separation on waste ore to produce a tantalite concentrate by-product as well as improving spodumene recovery.

In the final concentrate product, the moisture content must be regulated. This serves two purposes. The first is to avoid fluidisation during transport and the second is to reduce the overall weight of the material, leading to lower transport and storage costs. In warm climates, such as those in Australia, the heat from the environment is sufficient for this purpose and the concentrate is dried by solar evaporation in storage.

Stage 1: Lithium sulphate production

Concentrate processing will differ based on the final refined product. However, both carbonate and hydroxide production share the same initial stages of production:

- **Calcination:** The mineral feedstock is heated to approximately 1,050 °C inside a rotary kiln for approximately two hours. Calcining causes spodumene crystals to undergo a phase change from α to β, making the lithium in the mineral structure amenable to leaching with sulphuric acid.

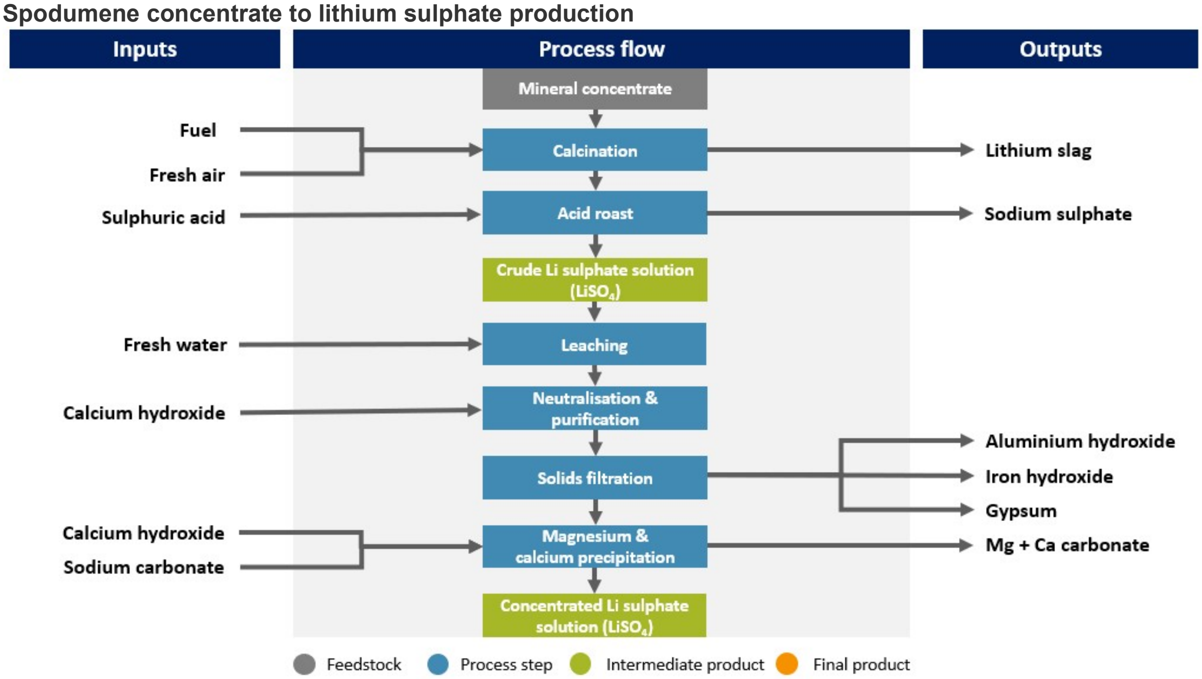
- **Acid roast:** The calcine is mixed with sulphuric acid in a rotary kiln and heated to 250 °C. Lithium sulphate is produced as the lithium present in the mineral reacts with sulphuric acid.

- **Leaching:** Water-soluble lithium sulphate is reacted with fresh water in leaching tanks and solid lithium sulphate, along with other metal impurities, forms a solution for further refining.

- **Neutralisation and impurity removal:** Calcium hydroxide is added to the lithium sulphate solution precipitating impurities (Al hydroxide, Fe hydroxide and gypsum) from the solution. Any solids present in the solution are removed via filtration.

- **Magnesium and calcium precipitation:** Magnesium and calcium are removed from the solution via the addition of calcium hydroxide and sodium carbonate, respectively; these precipitate out from the solution and are removed. The remaining concentrated liquor is sent on for further refining into either lithium carbonate or lithium hydroxide-monohydrate.

Spodumene concentrate to lithium sulphate production



Source: Wood Mackenzie

While spodumene is the predominant mineral feedstock for lithium carbonate and hydroxide production, lepidolite is also commonly used. As lepidolite is a mica mineral, it contains larger amounts of impurities within its structure. However, the production pathways between the two minerals only differ significantly up to the production of a concentrated lithium sulphate solution. Once this is produced, further production of refined lithium carbonate or hydroxide is the same for both feedstocks (with minor steps to remove further impurities in the lepidolite pathway).

There are two currently operational methods producing lithium sulphate solution from mica concentrate. Both rely on the beneficiation of lepidolite ore to produce a ground concentrate, with different steps following. The first method adds sulphuric acid to the ground mica concentrate and heats it to 250 °C, whereas the other adds potassium sulphate to the mica concentrate and heats it to 850 °C. The result in both cases is the production of a crude soluble lithium sulphate, though with additional impurities present in the solution as sulphates. Once this step has been completed, the solution is processed in largely the same method as spodumene concentrate, following with leaching and further steps in the flowsheet. If the lepidolite concentrate has been treated with potassium sulphate, during magnesium and calcium precipitation excess potassium sulphate must be removed from the solution. This is done via heating and evaporation, allowing the potassium sulphate to crystallise out of solution.

Stage 2: Refined lithium production

After this point, the flowsheets for hydroxide and carbonate split. Production of lithium carbonate involves the following:

- **Ion exchange**: The ion exchange functions as a further purification stage prior to the conversion of lithium sulphate into a carbonate product. The ion exchange uses a resin washed over the solution to selectively remove further calcium and magnesium within the mother liquor. However, depending on if the plant is aiming to produce a carbonate or hydroxide product, the chemical makeup of the resin used will vary.

- **Crude lithium carbonate crystallisation**: This segment produces a technical-grade carbonate product. The lithium sulphate solution is heated, and sodium carbonate added. Lithium sulphate reacts with the sodium carbonate and produces a low-solubility lithium carbonate that crystallises. The remaining sulphate reacts with sodium forming sodium sulphate. This is then pumped to the sodium sulphate crystalliser for removal. The remaining solution is washed and centrifuge and dewatered. If an operation is producing a technical-grade product, this is then dried in a rotary dryer and packaged for sale. If producing a refined battery-grade product, the solution is pumped to the bicarbonate process.

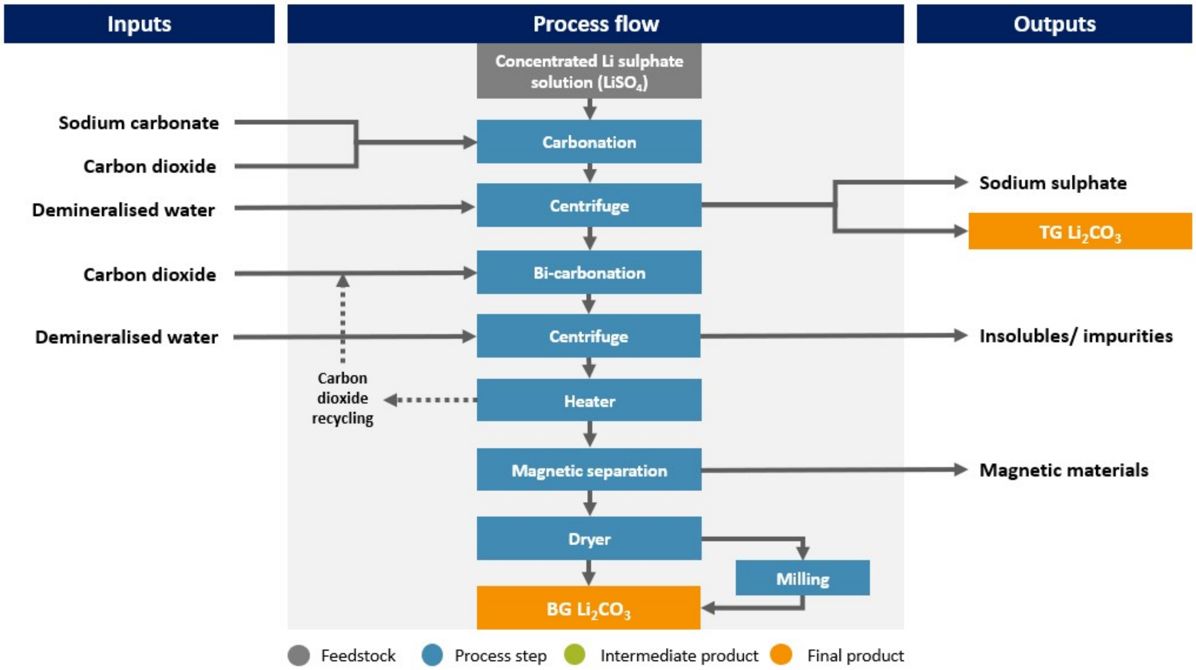
- **Sodium sulphate crystallisation**: The spent liquor from the crude lithium carbonate crystallisation stage has a small volume of sulphuric acid added to the solution to convert remaining lithium carbonate to lithium sulphate. Using a forced evaporative crystalliser, the sodium sulphate is crystallised out of solution under temperature-controlled conditions to produce an anhydrous form of sodium sulphate.

- **Sodium sulphate drying**: The sodium sulphate is dewatered and dried in a rotary dryer. The sodium sulphate product is then discharged, cooled and bagged for sale.

- **Pure lithium carbonate crystallisation**: The crude lithium carbonate is re-pulped and dissolved in fresh water. The solution is then sparged with carbon dioxide, which reacts with lithium carbonate to form a soluble lithium bicarbonate. During this stage, further sodium and sulphate is removed during the sparging process. In addition, calcium and magnesium are removed via filtration and ion exchange. The liquor is then heated, and lithium bicarbonate decomposes into lithium carbonate, these crystals are then washed and dewatered. After the production of lithium carbonate, the slurry mixture is passed through a magnetic separator to remove any magnetic material. This magnetic material is mainly from the production process as opposed to material present within the concentrate feedstock.

- **Drying and packaging**: The lithium crystals are dried in a rotary dryer before being packaged for sale.

Lithium sulphate to lithium carbonate production flowsheet



Source: Wood Mackenzie

Production of lithium hydroxide involves the following steps after production of a lithium sulphate liquor:

- **Ion exchange**: The ion exchange functions as a further purification stage prior to the conversion of lithium sulphate into a hydroxide product. The ion exchange uses a resin washed over the solution to selectively remove further calcium and magnesium within the mother liquor. However, depending on if the plant is aiming to produce a carbonate or hydroxide product, the chemical makeup of the resin used will vary.

- **Solution causticisation**: Sodium hydroxide is added to the mother liquor to convert the contained lithium sulphate to lithium hydroxide. This also produces sodium sulphate as a by-product.

- **Sodium sulphate decahydrate crystallisation**: The solution is refrigerated using a heat exchanger. This liquor contains small amounts of lithium hydroxide and sodium sulphate and water is added to balance the salt concentration to precipitate the sodium sulphate out of solution. This solid cake of sodium sulphate decahydrate is then separated via centrifuging and the solids are washed in cold water. This wash is recycled back to the crystalliser.

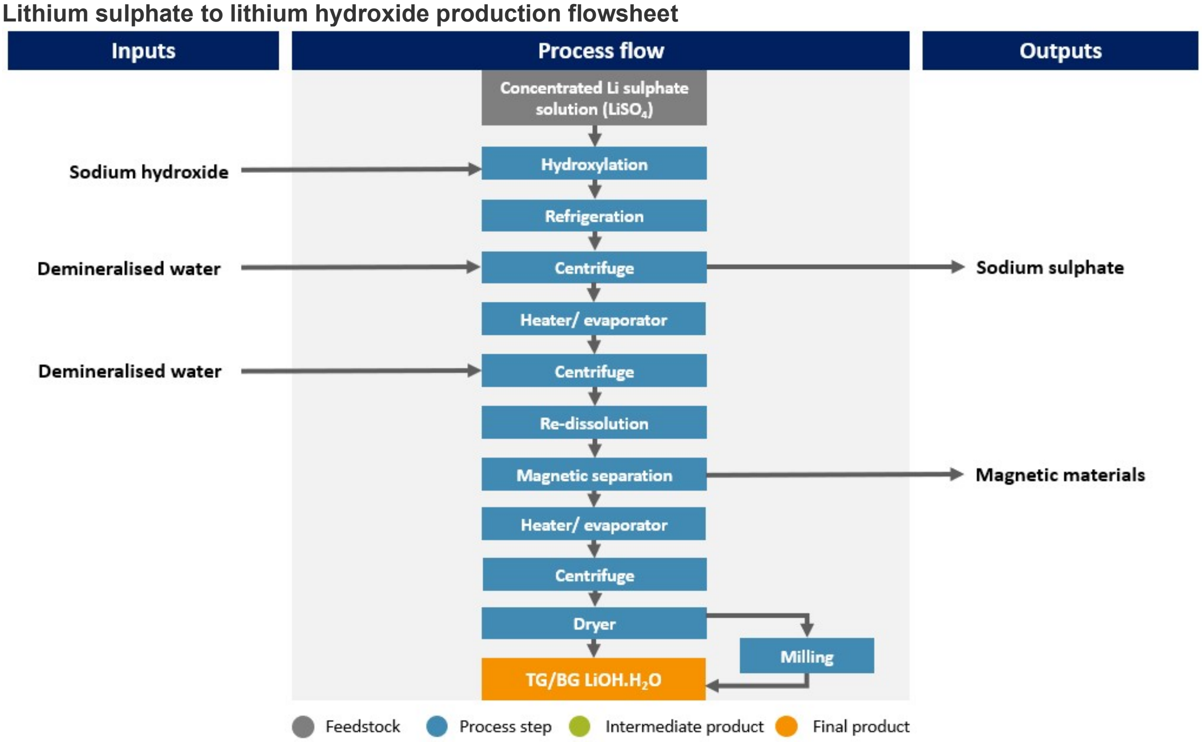
- **Sodium sulphate drying and packaging**: Sodium sulphate decahydrate is dried in a kiln to produce an anhydrous form of sodium sulphate. The crystals are then heated, and the excess water and moisture is removed. The sodium sulphate is discharged from the dryer, cooled and bagged for sale.

- **Crude lithium monohydrate evaporative crystallisation**: The remaining solution is heated, and water evaporated to increase the concentration of lithium hydroxide in solution and bring it close to saturation point. Lithium hydroxide monohydrate is then crystallised and this discharge is routed to a centrifuge for separation from the liquor. This produces a technical-grade lithium hydroxide product, which can then be dried and packaged for sale.

- **Pure lithium monohydrate evaporative crystallisation**: The crystallised lithium hydroxide monohydrate is dissolved in de-ionised water and recycled mother liquor (which is recycled at the end of this stage) and re-crystallised in an evaporator. The crystals are then filtered and washed via a centrifuge with more de-ionised water to remove the process liquor. The separated liquor is recycled for use in earlier stages of the plant. After the production of lithium hydroxide, the slurry mixture is passed through a magnetic separator. This magnetic material is mainly from the production process as opposed to material present within the concentrate feedstock.

- **Drying and packaging**: The lithium hydroxide crystals are dried in a fluidised bed dryer. This process is temperature controlled to prevent the lithium hydroxide monohydrate from becoming dehydrated to its anhydrous form. The lithium hydroxide monohydrate crystals are left to cool and packaged for sale.

Lithium sulphate to lithium hydroxide production flowsheet



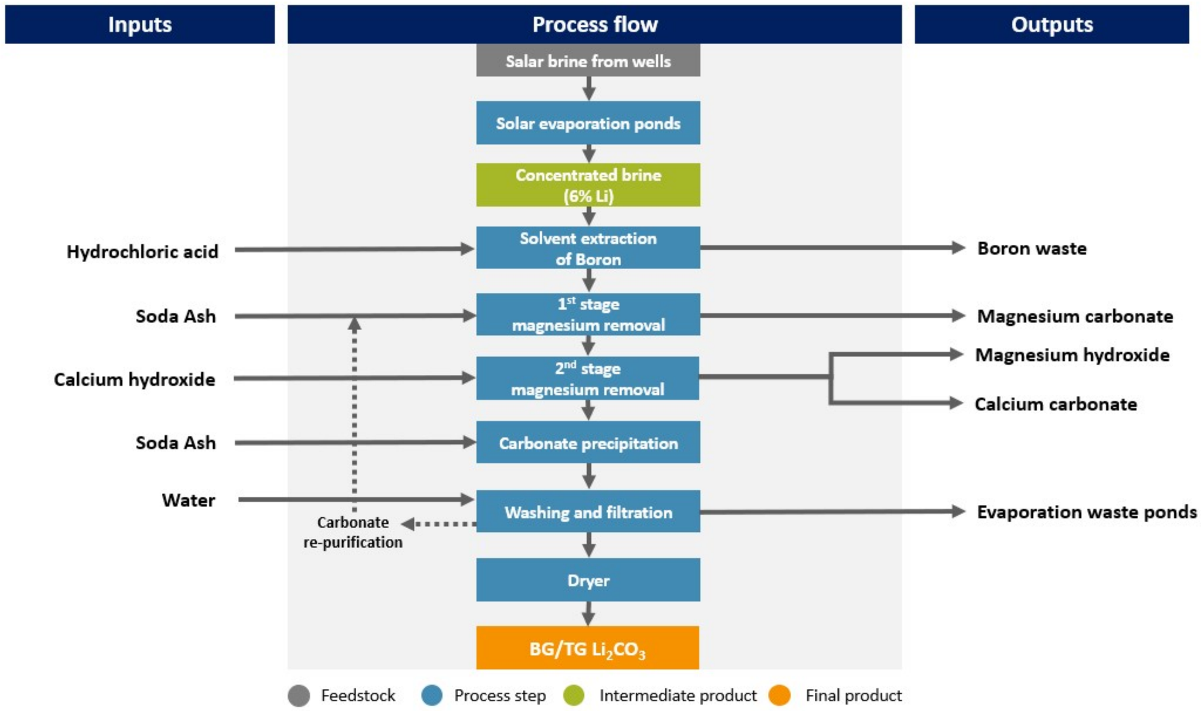
Source: Wood Mackenzie

Brine processing

Once a lithium chloride product has been produced, both DLE and evaporation flowsheets purify and concentrate this to a refined lithium carbonate product.

SQM and Albemarle use very similar pathways for lithium carbonate production at their Salar de Atacama operations and through their respective integrated Salar de Carmen and La Negra facilities. This breakdown of a solar evaporation route is based off information known about their processing methodology. In the solar evaporation pathway, the 6% Li content lithium chloride undergoes solvent extraction with hydrochloric acid, along with smaller quantities of sulphuric acid and caustic soda, to remove boron present in solution. After this, magnesium is removed in a primary and secondary stage. In the primary stage, the waste liquor from the final stage is redirected into the solution along with additional sodium carbonate (soda ash). The waste liquor contains residual sodium carbonate and this, along with the additional sodium carbonate, precipitates magnesium as magnesium carbonate. The solution is then pumped to the secondary stage and calcium hydroxide (milk of lime) is added to precipitate magnesium in the form of magnesium hydroxide. The calcium in solution then precipitates out as calcium carbonate. Both magnesium hydroxide and calcium carbonate are removed via filtering. The purified solution is then sent on to the lithium carbonate stage. To produce a lithium carbonate product, sodium carbonate is added to the solution under elevated temperatures of approximately 60 °C to 70 °C. Lithium carbonate is precipitated out of solution and the crystallised product is washed with fresh water and dried and packaged for sale. At this point, the waste liquor is recycled and redirected to the primary magnesium removal stage.

Brine to lithium carbonate production via solar evaporation – using SQM's and Albemarle's production pathways as case studies



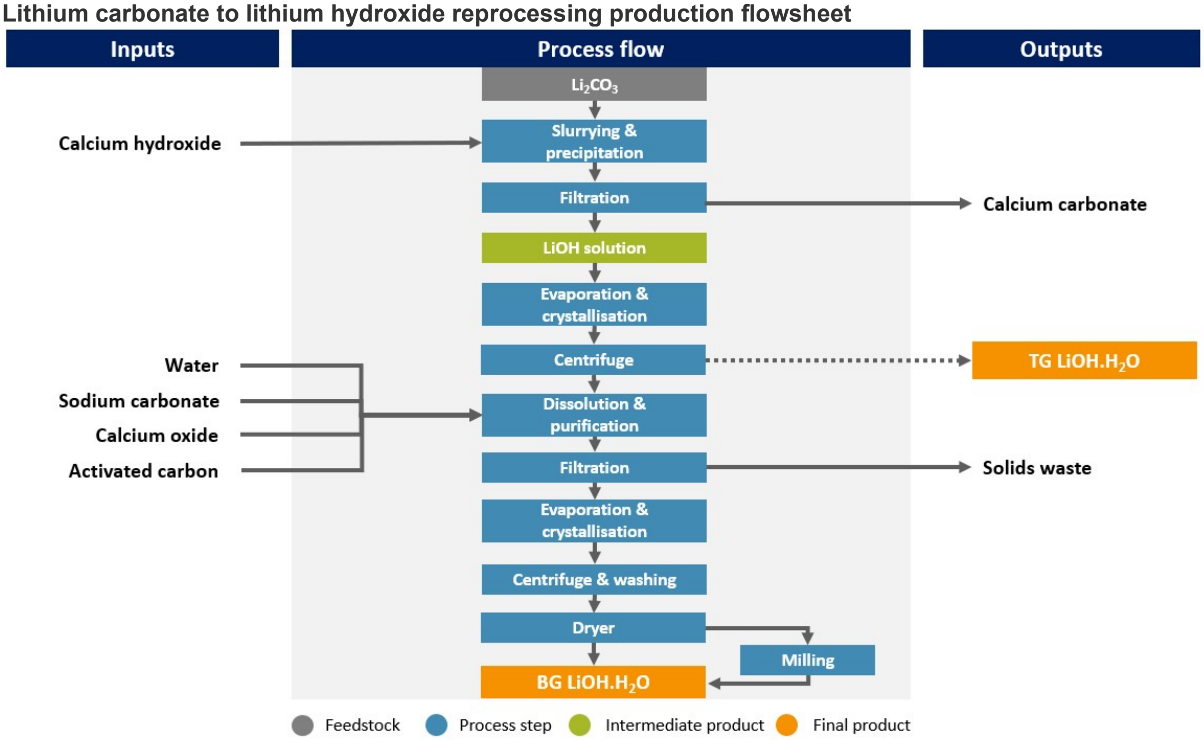
Source: Wood Mackenzie

The DLE production pathway examined in this section is based on information known about Livent's Salar del Hombre Muerto operation in Argentina and its integrated Güemes plant. Magnesium and calcium present in the brine solution are removed with sodium hydroxide (caustic soda) and sodium carbonate. These cause magnesium and calcium to precipitate out of solution as magnesium hydroxide and calcium carbonate, similar to the evaporative process. Boron in solution is removed via ion exchange and the remaining purified solution is heated and sodium carbonate is added. In this final stage, lithium precipitates out as lithium carbonate, the crystals are then centrifuged, and the product is dried and packaged for sale.

Re-processing: Lithium carbonate to lithium hydroxide

The refining of a hydroxide product from carbonate feedstock uses calcium hydroxide (lime). The lithium carbonate slurry is heated, and calcium hydroxide is added. If there is an imbalance of lithium carbonate and calcium hydroxide, then an excess of either product will be found in the final mixture. When heated and stirred, the lithium carbonate dissolves and calcium carbonate precipitates out. The slurry is filtered to separate the calcium carbonate from the lithium hydroxide still in solution. The slurry then undergoes heating and evaporation to crystallise the lithium hydroxide monohydrate, which is then separated via centrifuge. This produces a technical-grade lithium hydroxide monohydrate, which can be dried and packaged for sale. If a battery-grade quality product is needed, this form of lithium hydroxide is then re-dissolved in fresh water with the remaining impurities removed via the addition of sodium carbonate, calcium oxide and activated carbon. Filtration is then used to remove the solids and a battery-grade lithium hydroxide product crystallises from solution. This is then centrifuged and washed, dried and packaged for sale.

Lithium carbonate to lithium hydroxide reprocessing production flowsheet



Source: Wood Mackenzie